Based on [RG + MLHAD; 2602.01509]

HD Sense

An efficient method for ranking observable sensitivity

Rikab Gambhir

gambhirb@ucmail.uc.edu

In Collaboration with *MLHAD*: Benoît Assi, Christian Bierlich, Phil Ilten, Tony Menzo, Stephen Mrenna, Manuel Szewc, Michael K. Wilkinson, and Jure Zupan

LPCC MCWG Tuning Forum

Fundamental Questions

1. "What is the best set of observables for constraining the parameters of a model?"

2. "How can we determine this best set *cheaply*?"

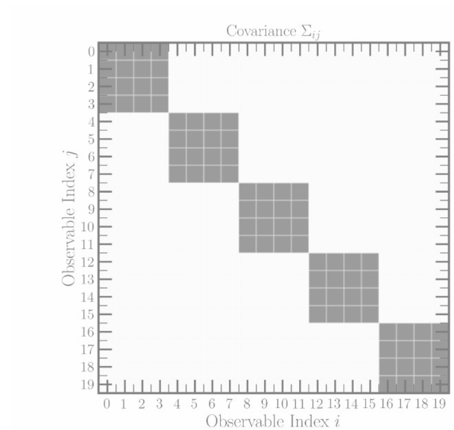
These two questions motivate the *HD Sense score*.

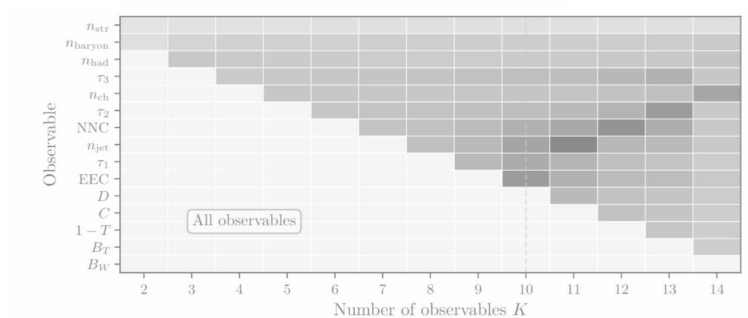
Outline

1. The HD Sense Score

Defining and explaining our metric

$$\mathcal{S}_{\mathrm{HD}}(\mathcal{X}) \;=\; \mathrm{Info}(\mathcal{X})\left[1 - \beta\,\mathcal{P}_{\mathrm{overlap}}(\mathcal{X})\right]$$





2. Motivation and Toys

Theory explanation and Gaussian Example

3. Physics Example

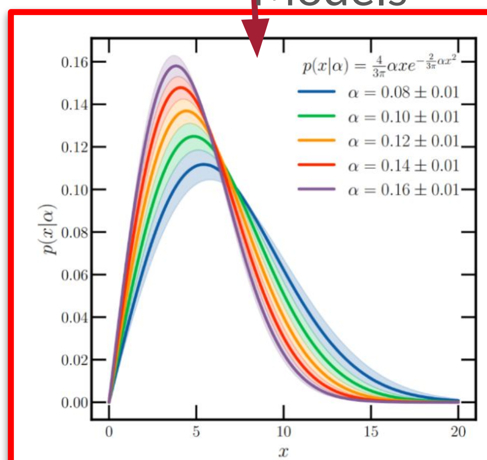
PYTHIA and Hadronization

Problem Statement

$$\boldsymbol{\theta} = \{\log a_{\mathrm{Lund}}, \log b_{\mathrm{Lund}}, \log(\sigma_{p_T}/\mathrm{GeV}), \log \rho, \log \xi\}$$

Parameterizes

Models



Constrains

We have a complex parameterized model with some hard-to-estimate likelihood

$$\mathcal{L}(\boldsymbol{\theta}|\mathcal{O}_1, \ldots, \mathcal{O}_K)$$

We would like to determine which subset of these observables constrains θ the most.

Standard statistics (e.g. Neyman-Pearson) tells us how to do this if we have the likelihood.

But the likelihood is hard and expensive to estimate — what can we do if we only have access to marginal distributions?

The HD Sense Score

Let *X* be a set of observables. Assume we have access to the *marginal* distributions of each observable, but not necessarily any correlations. We define the **HD[*] Sense Score**:

$$\boxed{\mathcal{S}_{\mathrm{HD}}(\mathcal{X}) \; = \; \mathrm{Info}(\mathcal{X}) \left[1 - \beta\, \mathcal{P}_{\mathrm{overlap}}(\mathcal{X}) \right]}$$

This is a single scalar that assigns a score to a subset of observables.

To pick a set of observables: calculate the score across many subsets, and pick the best!

[*] HD stands for "High Dimension"

The HD Sense Score: The Info Term

Let *X* be a set of observables. Assume we have access to the *marginal* distributions of each observable, but not necessarily any correlations. We define the **HD Sense Score**:

$$\mathcal{S}_{\mathrm{HD}}(\mathcal{X}) = \boxed{\mathrm{Info}(\mathcal{X})} \left[1 - \beta\, \mathcal{P}_{\mathrm{overlap}}(\mathcal{X})\right]$$

bservables.

ny subsets,

$$\mathrm{Info}(\mathcal{X}) = \sum_{i \in \mathcal{X}} \mathrm{Tr}\, I^{(i)}$$

Trace of the Fisher Info of each individual observable

The more informative each individual observable is, the higher this term!

$$I^{(i)}_{ab}(\boldsymbol{\theta}) = \mathbb{E}_{p_i(\mathcal{O}_i|\boldsymbol{\theta})}\left[\frac{\partial \log p_i(\mathcal{O}_i|\boldsymbol{\theta})}{\partial \theta_a} \frac{\partial \log p_i(\mathcal{O}_i|\boldsymbol{\theta})}{\partial \theta_b}\right]$$

Fisher Info: Calculable from 1D Histograms

The HD Sense Score: The Overlap Term

Let X be a set of observables. Assume we have access to the *marginal* distributions of each observable, but not necessarily any correlations. We define the **HD Sense Score**:

$$\mathcal{S}_{\mathrm{HD}}(\mathcal{X}) = \mathrm{Info}(\mathcal{X})\left[1 - \beta\, \boxed{\mathcal{P}_{\mathrm{overlap}}(\mathcal{X})}\right]$$

$$\mathrm{Info}(\mathcal{X}) = \sum_{i\in\ldots}$$

The more inform…

$$I^{(i)}_{ab}(\theta) = \mathbb{E}_{p_i(\mathcal{O}_i|\theta)}\left[\ldots\right.$$

$$\mathcal{P}_{\mathrm{overlap}}(\mathcal{X}) = \frac{2}{\sum_{k\in\mathcal{X}} \mathrm{Tr}[I^{(k)}]} \sum_{\substack{i,j\in\mathcal{X}\\ i<j}} \sqrt{\mathrm{Tr}[I^{(i)}]\mathrm{Tr}[I^{(j)}]\cos(\Phi^F_{ij})},$$

Penalize redundant observables in the score — accounts for potential correlations between observables

Frobenius Inner Products of Fisher Infos

$$\cos(\Phi^F_{ij}) = \frac{\langle I^{(i)}, I^{(j)}\rangle_F}{||I^{(i)}||_F\, ||I^{(j)}||_F}$$

The HD Sense Score: The Penalty Strength

Let *X* be a set of observables. Assume we have access to the *marginal* distributions of each observable, but not necessarily any correlations. We define the **HD Sense Score**:

$$\mathcal{S}_{\mathrm{HD}}(\mathcal{X}) = \mathrm{Info}(\mathcal{X}) \left[1 - \boxed{\beta}\, \mathcal{P}_{\mathrm{overlap}}(\mathcal{X}) \right]$$

A single hyperparameter controlling how redundancy should be penalized — related to the average (unknown) correlation of observables.

Based on scaling arguments, a reasonable **Heuristic**

$$\beta = \frac{\beta_0}{\max_{\mathcal{X}} \mathcal{P}_{\mathrm{overlap}}(\mathcal{X})} \quad \text{with } \beta_0 = 0.5$$

[Aside] Fisher Information on Binned Data

To use the HD Sense Score, we need to be able to estimate 1D Fisher Information matrices. Given binned data, we can estimate the Fisher Information with multinomials:

Fisher Info of observable factorizes into (Fisher of bin values) x (Derivatives)

$$I^{(i)}_{ab}(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{n}\sim p(\boldsymbol{n}|\boldsymbol{\alpha}(\boldsymbol{\theta}))}\left[\frac{\partial}{\partial\theta_a}\log p\big(\boldsymbol{n}|\boldsymbol{\alpha}(\boldsymbol{\theta})\big)\frac{\partial}{\partial\theta_b}\log p\big(\boldsymbol{n}|\boldsymbol{\alpha}(\boldsymbol{\theta})\big)\right]$$

$$= \sum_{m,n=1}^{B-1}\frac{\partial\alpha_m}{\partial\theta_a}\frac{\partial\alpha_n}{\partial\theta_b}\tilde{I}_{mn}(\boldsymbol{\alpha}),$$

Estimated using reweighting + linear fits

Fisher Info of bin values α with counts *n*.

$$\tilde{I}_{mn}(\boldsymbol{\alpha}) = \mathbb{E}_{\boldsymbol{n}\sim p(\boldsymbol{n}|\boldsymbol{\alpha})}\left[\left(\frac{n_m}{\alpha_m} - \frac{n_B}{1-\sum\alpha_p}\right)\left(\frac{n_n}{\alpha_n} - \frac{n_B}{1-\sum\alpha_p}\right)\right]$$

$$= N\left(\delta_{mn}\frac{1}{\alpha_m} + \frac{1}{1-\sum\alpha_p}\right),$$

Using the HD Sense Score

All of this has been conveniently implemented!

A PYTHIA plug-in library, consisting of a simple UserHook. Just requires linking to PYTHIA and FASTJET (details and instructions in readme).

These formulae are also simple enough to implement by hand in other non-PYTHIA contexts.

https://gitlab.com/pythia8-contrib/packages/hdsense

MLHAD University of CINCINNATI

Outline

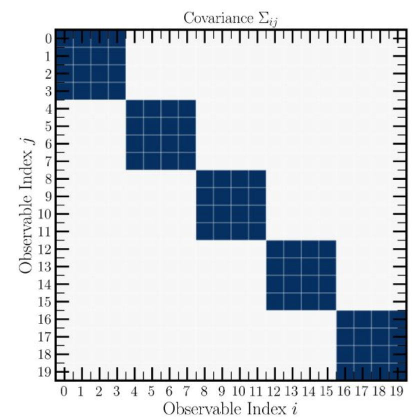
1. The HD Sense Score

Defining and explaining our metric

$$\mathcal{S}_{\rm HD}(\mathcal{X}) = \text{Info}(\mathcal{X})\left[1-\beta\,\mathcal{P}_{\rm overlap}(\mathcal{X})\right]$$

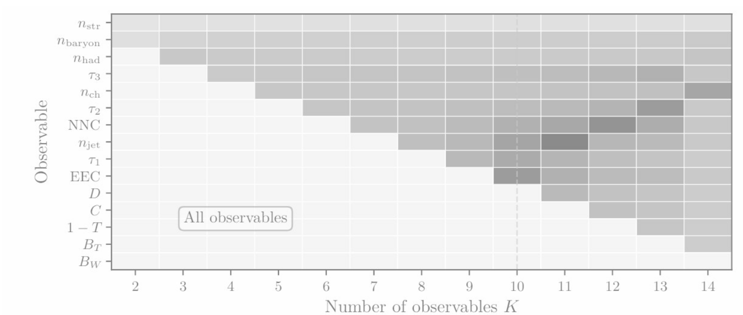
2. **Motivation and Toys**

Theory explanation and Gaussian Example



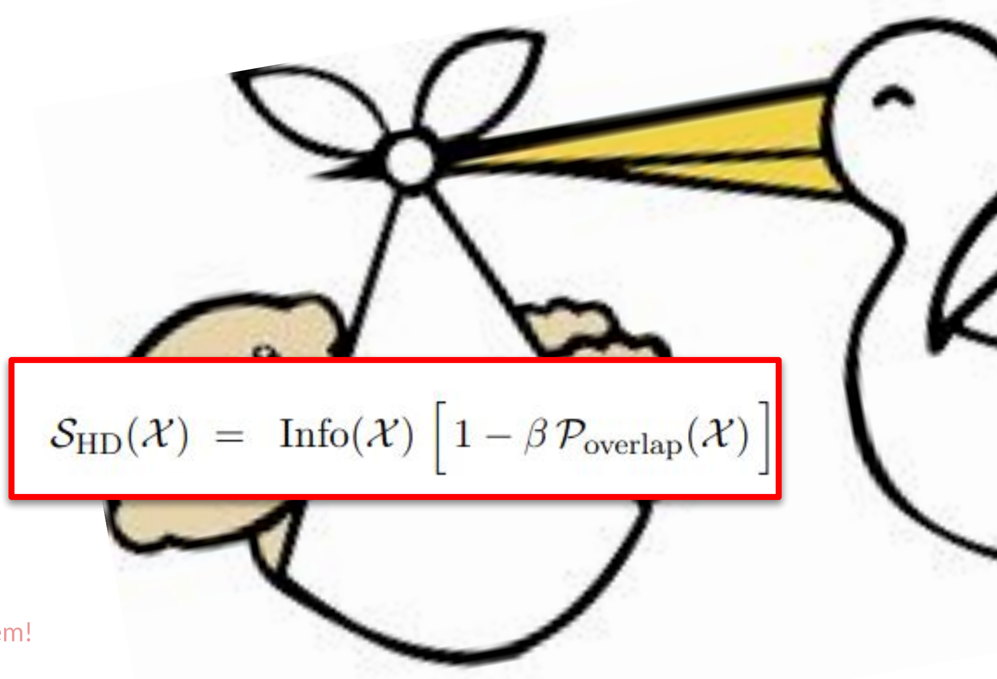
3. Physics Example

PYTHIA and Hadronization



“Where did this even come from?”

The HD Sense Score approximates the trace of an “effective” Fisher Information, where unknown correlation information has been “profiled” out.



$$\mathcal{S}_{\mathrm{HD}}(\mathcal{X}) \;=\; \mathrm{Info}(\mathcal{X})\left[1-\beta\,\mathcal{P}_{\mathrm{overlap}}(\mathcal{X})\right]$$

$$-2\mathcal{L}(\boldsymbol{\theta}|\mathcal{O}_1,\mathcal{O}_2,\ldots,\mathcal{O}_{N_{\mathrm{obs}}}) \approx$$

$$[\mathrm{constant}] + (\theta-\hat{\theta})_a I_{ab}(\theta-\hat{\theta})_b$$

The *full* Fisher Info: Everything about our problem!

Our Strategy: Start with the *exact*, *full* Fisher information, then systematically insert our ignorance about correlations. We will arrive at the HD Sense Score!

For more detail, see our full derivation in [**RG** + MLHAD; 2602.01509]

Profiling Out Correlations

If we have parameters η controlling how correlated observables are, we can separate them out from our physical observables θ.

$$I_{\text{pr.}}(\boldsymbol{\theta}) = I_{\theta\theta}(\boldsymbol{\theta}, \hat{\boldsymbol{\eta}}) - I_{\theta\eta}(\boldsymbol{\theta}, \hat{\boldsymbol{\eta}}) I_{\eta\eta}^{-1}(\boldsymbol{\theta}, \hat{\boldsymbol{\eta}}) I_{\eta\theta}(\boldsymbol{\theta}, \hat{\boldsymbol{\eta}})$$

General Gaussianity / Central Limit Theorem Assumptions, only assuming 2-point correlations

All correlation structure captured in ϱ

$$\mathcal{S}_{\text{pr.}}(\mathcal{X}) = \left(\sum_i \text{Tr}[I^{(i)}](\rho^{-1})_{ii}\right)\left(1 + 2\frac{\sum_{i<j}\sqrt{\text{Tr}[I^{(i)}]\,\text{Tr}[I^{(j)}]}\cos(\Phi_{ij})[\rho^{-1}]_{ij}}{\sum_i \text{Tr}[I^{(i)}](\rho^{-1})_{ii}}\right)$$

MLHAD University of CINCINNATI

Parameterizing our Ignornace

We do not know the correlation matrix ϱ. We must use inequalities to estimate it

$$\mathcal{S}_{\text{pr.}}(\mathcal{X}) = \left(\sum_i \text{Tr}[I^{(i)}](\rho^{-1})_{ii}\right)\left(1 + 2\frac{\sum_{i<j}\sqrt{\text{Tr}[I^{(i)}]\,\text{Tr}[I^{(j)}]}\cos(\Phi_{ij})[\rho^{-1}]_{ij}}{\sum_i \text{Tr}[I^{(i)}](\rho^{-1})_{ii}}\right)$$

β is related to the max value of the entries of ϱ^{-1}, which can be any value greater than zero.

$$\mathcal{S}_{\text{pr.}} \gtrsim \left(\sum_i \text{Tr}[I^{(i)}]\right)\left(1 - 2\beta\frac{\sum_{i<j}\sqrt{\text{Tr}[I^{(i)}]\,\text{Tr}[I^{(j)}]\cos(\Phi^F_{ij})}}{\sum_i \text{Tr}[I^{(i)}]}\right) \equiv \mathcal{S}_{\text{HD}}$$

Therefore, the HD Sense Score is an *approximate lower bound* for the trace of the full Fisher Info. This is why it is a good estimate for how much information is in a set of observables!

Gaussian Toy Example

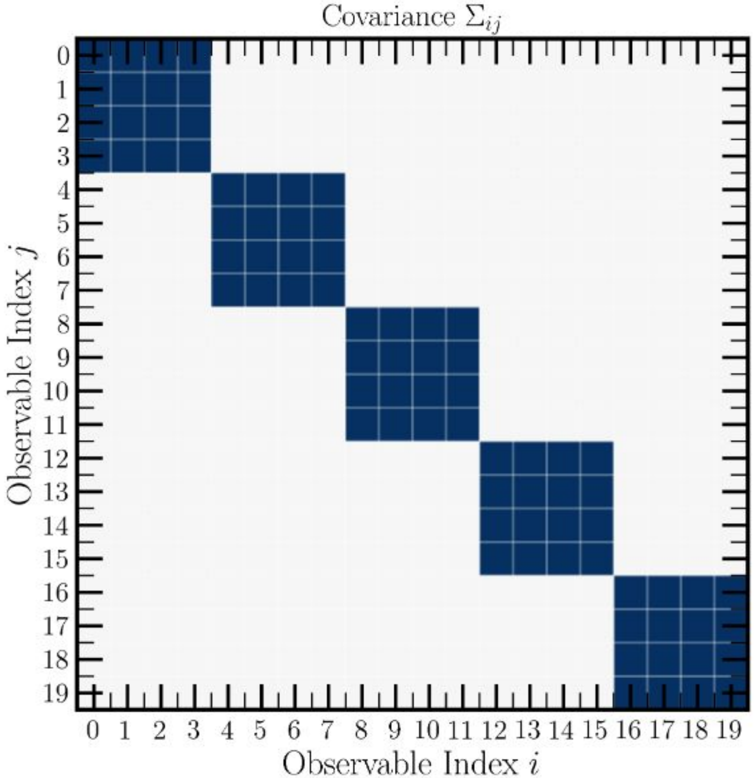
Let's play with toys!

If everything I promised is true, the HD Sense Score should remove *perfectly redundant observables.*

Our toy: 5 different uncorrelated observables, but perfectly copied 4 times each, as shown **to the right**.

$$p(\mathcal{O}_1, \ldots \mathcal{O}_K) \propto e^{-\frac{1}{2}(\mathcal{O}-\mu(\boldsymbol{\theta}))_i \Sigma^{-1}_{ij} (\mathcal{O}-\mu(\boldsymbol{\theta}))_j}$$

Pictured: Block Diagonal Covariance. Observables 0-3 are copies, 4-7 are copies, and so on



Prediction: For K = 5, the score should pick exactly 1 observable from each block.

Gaussian Toy Example: Results

	Penalty β	Selected observables	$\mathcal{S}_{\mathrm{HD}}$ score	Quality
	-0.1	$\{0_0, 1_0, 2_0, 3_0, 16_4\}$	6.749	Maximally subopt.
	0.0	$\{0_0, 1_0, 9_2, 10_2, 11_2\}$	5.000	Suboptimal
	0.1	$\{0_0, 4_1, 8_2, 12_3, 16_4\}$	4.234	Optimal
Heuristic Choice	**0.143**	$\boldsymbol{\{0_0, 4_1, 8_2, 12_3, 16_4\}}$	**3.960**	**Optimal**
	1.0	$\{0_0, 4_1, 8_2, 12_3, 16_4\}$	-2.654	Optimal

For □ > 0, we successfully choose one observable per block:
Redundant observables have been eliminated!

Conclusion: The HD Sense Score works as advertised!

[Aside] Curiosities

	Penalty β	Selected observables	S_{HD} score	Quality
	-0.1	$\{0_0, 1_0, 2_0, 3_0, 16_4\}$	6.749	Maximally subopt.
	0.0	$\{0_0, 1_0, 9_2, 10_2, 11_2\}$	5.000	Suboptimal
	0.1	$\{0_0, 4_1, 8_2, 12_3, 16_4\}$	4.234	Optimal
Heuristic Choice	**0.143**	$\boldsymbol{\{0_0, 4_1, 8_2, 12_3, 16_4\}}$	**3.960**	**Optimal**
	1.0	$\{0_0, 4_1, 8_2, 12_3, 16_4\}$	-2.654	Optimal

For □ < 0, correlations are *rewarded*, so we choose the *worst possible* observable set.
For □ = 0, correlations are *ignored*, so we get a random observable set.

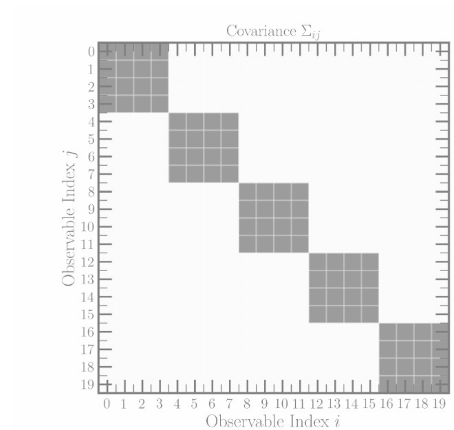
We rigged this example such that the trace of the Fisher Info is exactly 5.0. We underestimate it, *as advertised!*

Outline

1. The HD Sense Score

Defining and explaining our metric

$$\mathcal{S}_{\rm HD}(\mathcal{X}) = \mathrm{Info}(\mathcal{X})\left[1 - \beta\,\mathcal{P}_{\rm overlap}(\mathcal{X})\right]$$

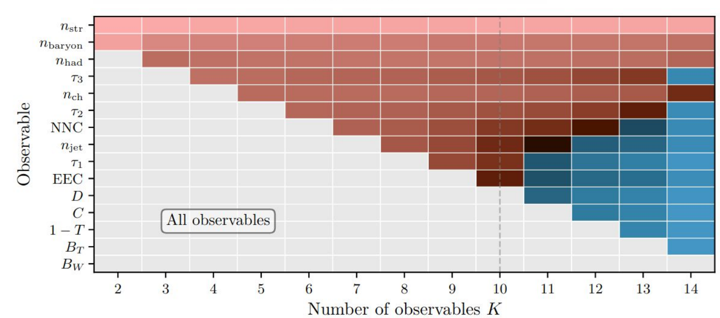


2. Motivation and Toys

Theory explanation and Gaussian Example

3. **Physics Example**

PYTHIA and Hadronization



Lund Strings

The specific example we (and likely, you) care about is **Hadronization**.

We would like to choose a subset of observables:

- *N_{Had}*
- *$N_{Strange}$*
- *$N_{Charged}$*
- *N_{Baryon}*
- *Thrust*
- *Total Broadening*
- *Wide Jet Broadening*
- *C-Parameter*
- *D-Parameter*
- *Number of Jets*
- *1-Jettiness*
- *2-Jettiness*
- *3-Jettiness*
- *Energy-Energy-Correlator*
- *Number-Number-Correlator*

To constrain the following 5 parameters of the Lund String Model:

$$\boldsymbol{\theta} = \{\log a_{\text{Lund}}, \log b_{\text{Lund}}, \log(\sigma_{p_T}/\text{GeV}), \log \rho, \log \xi\},$$

*We find it convenient to use log coordinates for all parameters

Setup and Events

In PYTHIA 8.3, we take the reference

$$\hat{\theta} = \log\{0.68, 0.98, 0.335, 0.217, 0.081\}$$

We generate 150 variations within a 5% radius of this point.

We calculate histograms for each observable. Bins are chosen such that the statistical error is no more than 1%. Gradients are estimated using linear fits.

We then apply the previously defined formulae and code!

Details

- 1000000 Events
- e^+e^- to quarks at $Q = M_Z$

Bin Choices:

HDSense:nAll:bins = {48,3.5,51.5}
HDSense:nStrange:bins = {16,-0.5,15.5}
HDSense:nCharge:bins = {19,3,41}
HDSense:nBaryon:bins = {8,-1,15}
HDSense:nJet:bins = {3,0.5,3.5}
HDSense:thrust:bins = {10,0.6,1.0}
HDSense:event_shape_C:bins = {10,0.0,1.0}
HDSense:event_shape_D:bins = {10,0.0,0.8}
HDSense:event_shape_BW:bins = {10,0.0,0.4}
HDSense:event_shape_BT:bins = {10,0.0,0.3}
HDSense:nSubjet1:bins = {10,0.0,0.7}
HDSense:nSubjet2:bins = {10,0.0,0.5}
HDSense:nSubjet3:bins = {10,0.0,0.4}

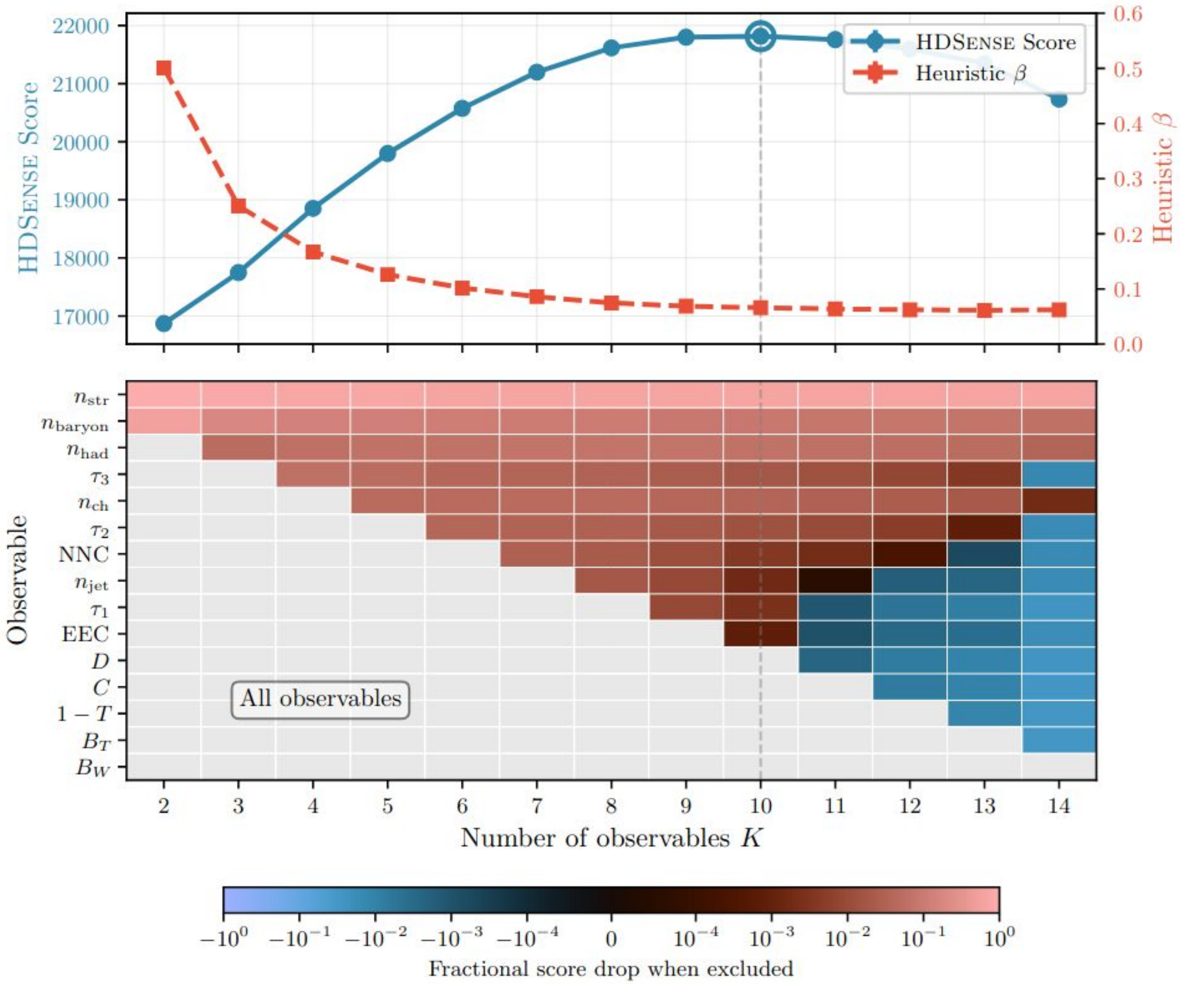
Results

We select for the best K observables of the 15.

We learn that multiplicities are the most constraining — makes sense!

IRC-safe shapes are the least constraining.

Diminishing returns after approximately 10 observables — many event shapes are redundant with each other.



MLHAD University of CINCINNATI

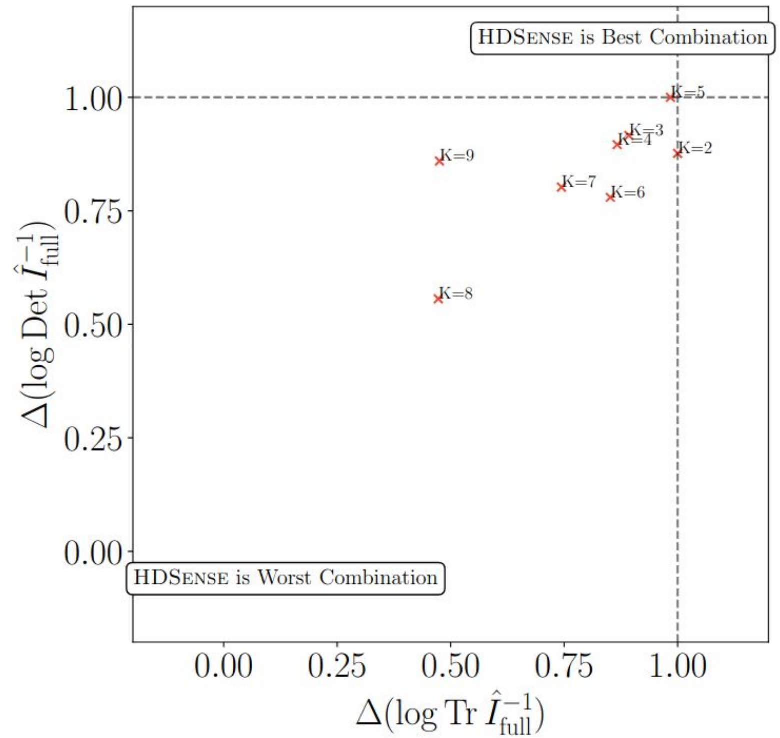
[Aside] Is HD Sense any good?

In the second part of this talk, we constructed HD Sense as an approximation to the full likelihood.

Just as a sanity check, let's use a relatively expensive approximation to the full likelihood — an **ML-learned BDT**

By Neyman Pearson, an ML classifier also learns the full likelihood. So we can use that to select observables.

But, for moderate *K*, the HD Sense Score picks out the best combination or close to it anyways!



[Aside] What does "Best Observables" Mean?

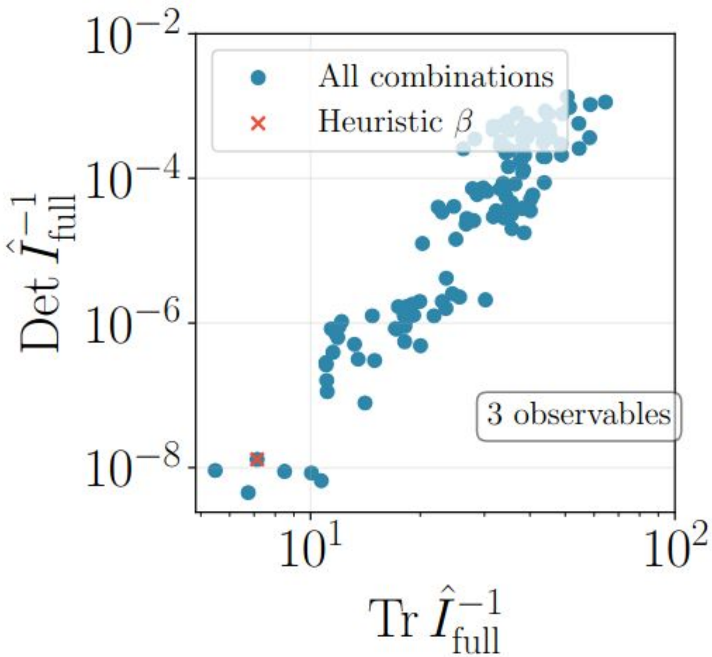
We took the HD Sense to be the *trace* of the Fisher info. By Cramer-Rao:

$$\mathrm{cov}(\hat{\boldsymbol{\theta}}) \geq I^{-1}(\boldsymbol{\theta})$$

Why not inverse-trace, or determinant, or any other scalar summary?

We could have, and they're all correlated. But trace is simplest and easiest, so we go with trace.

There is a rich literature on *optimal design*, which studies what aspects of the information to optimize, but this usually reduces to some combination of eigenvalues.



Based on [RG + MLHAD; 2602.01509]

Conclusion

The **HD Sense Score** is a well-motivated metric for cheaply determining the best observables to measure in the absence of information about correlations.

We have shown it works well for hadronization in PYTHIA. But it can be used equally well for any experimental design task.

Please email me your questions and ideas at *gambhirb@ucmail.uc.edu*!

Need help using the code? Let us know!

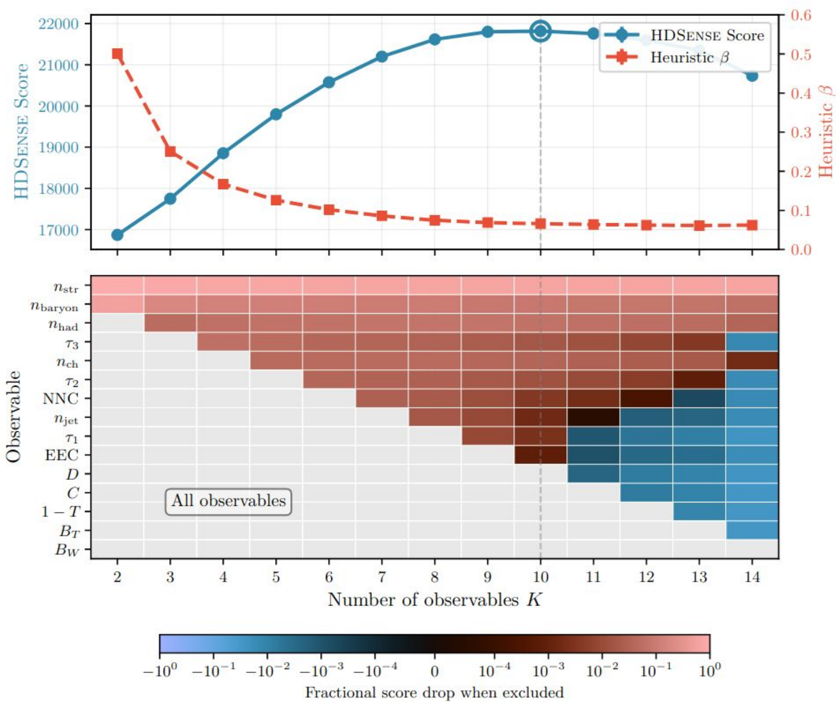
https://gitlab.com/pythia8-contrib/packages/hdsense

MLHAD University of CINCINNATI

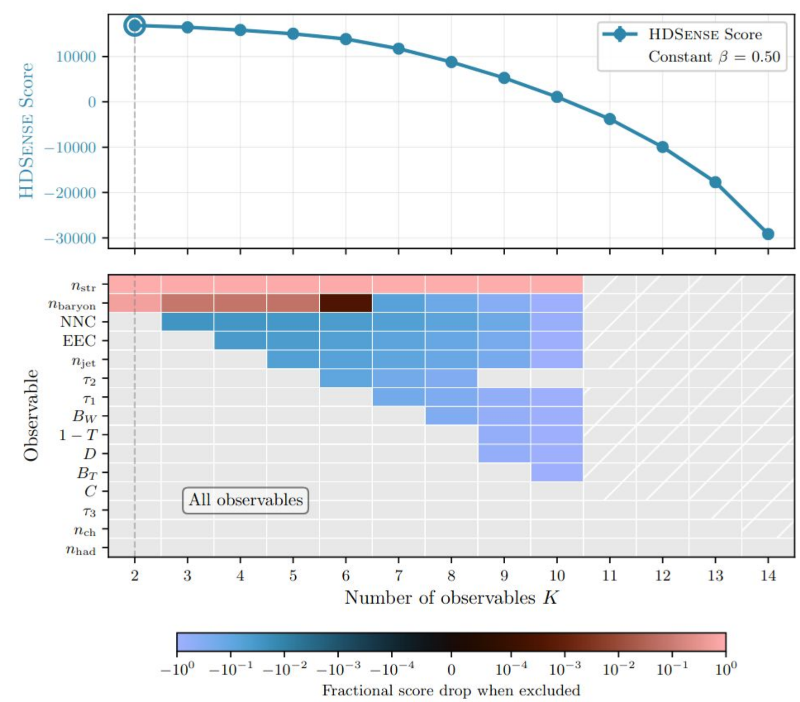
Appendices

Heuristic vs Fixed β

For Heuristic β, correlations scale naturally



For fixed β, very easy to over-estimate correlations



$$\beta = \frac{\beta_0}{\max_{\mathcal{X}} \mathcal{P}_{\mathrm{overlap}}(\mathcal{X})} \quad \text{with } \beta_0 = 0.5$$

Combining Experiments With Different Stats

Fisher info is additive between independent experiments, even with different observables!

Exp 1: High stats, but cannot resolve flavor (no n_{Baryon} *or* $n_{Strange}$)

Exp 2: 10x fewer stats, but can resolve flavor

We can analyze the joint experiment! n_{Had} gains relative importance vs. $n_{Strange}$, but $n_{Strange}$ is powerful enough to be selected even with low stats